

Integrali indefiniti immediati

$$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + c \quad \alpha \neq -1, \quad \int \frac{1}{\cos^2 x} dx = \operatorname{tg} x + c$$

$$\int \frac{1}{x} dx = \ln|x| + c \quad \alpha = -1 \quad \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + c$$

$$\int e^x dx = e^x + c, \quad \int \frac{-1}{\sqrt{1-x^2}} dx = \arccos x + c$$

$$\int \sin x dx = -\cos x + c, \quad \int \frac{1}{1+x^2} dx = \operatorname{arctg} x + c$$

$$\int \cos x dx = \sin x + c,$$

ESERCIZI n°1

$$\int \frac{\sin x}{\cos x} dx = - \int \frac{1}{\cos x} (-\sin x) dx = -\ln|\cos x| + c$$

ESERCIZIO n°2

$$\int \sin x \cos^2 x dx = - \int -\cos x \cos^2 x dx = \frac{\cos^3 x}{3} + c$$

ESERCIZIO n°3

$$\int \frac{1}{x^2 + x + 1} dx \quad N(x) < D(x)$$

$$\int \frac{1}{x^2 + 2\left(\frac{1}{2}\right)x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} dx = \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx = \frac{1}{\frac{3}{4}} \int \frac{1}{\left[\frac{4}{3}\left(x + \frac{1}{2}\right)^2\right] + 1} dx$$

QUADRATO DI UN BIENNIO
+ UNA COSTANTE

$$\Rightarrow \frac{4}{3} \int \frac{1}{\left[\frac{2(x+\frac{1}{2})}{\sqrt{2}}\right]^2 + 1} dx \Rightarrow \frac{4}{3} \int \frac{\sqrt{3}}{\left[\frac{2(x+\frac{1}{2})}{\sqrt{2}}\right]^2 + 1} dx = \frac{2}{\sqrt{3}} \cdot \operatorname{ARCTg} \cdot \frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right) + C$$